Tutorial 1 - Introduction to Microcontroller

Q1.1 Which of the following best describes a microcontroller?

(a) A microcontroller is any device that can provide intelligent control in a system.
(b) A microcontroller is a small computer on a single integrated circuit consisting of a CPU, clock, timers, IO ports and memory.
(c) A microcontroller is an integrated circuit with a fixed program running in it.
(d) A microcontroller is a micro-processor.

b

Q1.2 Which of the following components cannot be externally connected to a microcontroller?

(a) Crystal clock oscillator
(b) CPU
(c) RAM
(d) EEPROM

b

Q1.3 What is the correct sequence of a GCC build sequence?

(a) Compiler, Pre-Processer, Linker, Assembler
(b) Pre-Processor, Compiler, Assembler, Linker
(c) Linker, Compiler, Pre-Processor, Assembler
(d) Pre-Processor, Compiler, Linker, Assembler

b

Q1.4 Assuming Microcontroller A has a higher MIPS than Microcontroller B. What could be the possible result if same C code was compiled and executed on Microcontroller A and B?

(a) Executable binary size is larger for Microcontroller A than B
(b) Code executes faster on Microcontroller A than B
(c) Code executes faster on Microcontroller B than A
(d) Code requires more RAM on Microcontroller A than B

b

Q1.5 Which Arm® 32-bit RISC does core STM32F103RB incorporate?

(a) Cortex-M0
(b) Cortex-M0+
(c) Cortex-M3
(d) Cortex-M33

C

Q1.6 Why is inline assembly used with C/C++?

When tasks are complicated to do with C/C++, assembly is used to solve these problems.